

SYSTEM ADMIN WITH IT SERVICE MANAGEMENT & AI EMPHASIS

YEAR : 3rd YEAR

COLLEGE CODE : MSU220

COLLEGE NAME : Sri Sankara Bhagavathi Arts and Science College,
Kommadikottai

DEPARTMENT : B.Sc IT

SEMESTER : 5

TITLE : Script-Controlled ACL – Restrict Record Access Based on Field Value

GOOGLE DRIVE LINK :

GITHUB LINK : <https://github.com/manivannan26122006-hub/SmartBridge-ServiceNow-Project>

TEAM DETAILS

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Team Size : 4 Members

ABSTRACT

The project "Script-Controlled ACL – Restrict Record Access Based on Field Value" focuses on implementing secure record-level access control in the ServiceNow platform using Access Control Lists (ACLs). The project creates a

custom table named Institution Details, configures users and custom roles, and applies Script-Controlled Read, Create, Write, and Delete ACLs. The system allows only authorized users with the required roles to access EEE branch records, while administrators retain full access. This implementation improves data security, prevents unauthorized access, and demonstrates practical role-based access control in ServiceNow.

1. INTRODUCTION

ServiceNow is a cloud-based platform used for IT Service Management and enterprise application development. Organizations often store confidential information that should only be accessible to authorized users. Access Control Lists (ACLs) provide a secure method to control who can view, create, edit, or delete records.

This project implements Script-Controlled ACLs on the Institution Details table. The system evaluates both user roles and the Branch field to determine whether access should be granted. Users with the appropriate roles can perform their assigned operations, while administrators maintain complete control over all records.

This matches the same style and headings as your friend's report while using your project title and your team details.

Phase 1 – Brainstorming & Ideation

1. Introduction

Educational institutions store sensitive information such as student roll numbers, branch details, faculty information, email addresses, and phone numbers. This information should be accessible only to authorized users. ServiceNow provides Access Control Lists (ACLs) to protect records by checking user roles and record conditions before granting access.

This project demonstrates how to implement a Script-Controlled ACL that allows only authorized users to access records belonging to the EEE branch, while administrators retain full access to all records.

2. Problem Statement

In many organizations, all users should not have permission to view or modify every record. Without proper security, confidential student information can be accessed by unauthorized users.

The problem addressed in this project is to restrict record access based on both the Branch field value and user roles, ensuring that only eligible users can perform specific operations.

3. Project Idea

The project involves creating a custom table named Institution Details and configuring multiple ACLs for Read, Create, Write, and Delete operations. JavaScript scripting is used inside the Read ACL to check whether the logged-in user has the required role before allowing access.

4. Objectives

- Create a test user in ServiceNow.
- Create custom roles (bb1, bb2, bb3, bb4).
- Build the Institution Details custom table.
- Add student-related fields to the table.
- Configure Script-Controlled Read ACL.
- Configure Create, Write, and Delete ACLs.
- Test the security using impersonation.

5. Existing System

In an unsecured system:

- Any user may access sensitive records.

- Data privacy is not maintained.
- Unauthorized users may edit or delete information.

6. Proposed Solution

The proposed solution combines Role-Based Access Control (RBAC) with Script-Controlled ACLs. The system checks whether the logged-in user has the required role before granting access.

Access Rules

User	Access
Admin	Full access
bb1	View EEE records
bb2	Create records
bb3	Edit records
bb4	Delete records

7. Expected Benefits

- Protects sensitive student records.
- Prevents unauthorized access.
- Improves data security.
- Demonstrates practical implementation of ServiceNow ACLs.
- Supports secure record management.

8. Expected Outcome

After completing the project, the Institution Details application will successfully enforce record-level security. Administrators will have full control, users with bb1 will view only EEE records, and additional roles will control Create, Write, and

Delete permissions, making the application secure and compliant with role-based access requirements.

Phase 2 – Requirement Analysis

1. Introduction

Requirement Analysis is the process of identifying what the system needs to do before development begins. In this project, the requirements focus on creating a secure ServiceNow application that restricts access to student records based on user roles and the Branch field value.

2. Functional Requirements

The system must:

- Create a test user (EEE User).
- Create custom roles (bb1, bb2, bb3, bb4).
- Create the Institution Details custom table (u_institution_details).
- Add fields such as Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number, and Description.
- Create multiple records for ECE, EEE, and CSE branches.
- Configure Read, Create, Write, and Delete ACLs.
- Support impersonation testing.

3. Non-Functional Requirements

- Security: Only authorized users should access records.
- Performance: ACL checks should execute quickly.
- Reliability: Access permissions should work consistently.
- Usability: The application should be easy to use.

4. User Requirements

User	Requirement
------	-------------

Admin	Full access
bb1	View EEE records
bb2	Create records
bb3	Edit records
bb4	Delete records

5. System Requirements

- ServiceNow Developer Instance
- Admin access
- Web browser
- Internet connection

6. Table Requirements

Table Name: u_institution_details

Fields

Field	Type
Student Roll Number	Auto Number
Student Name	Reference (User)
Faculty Name	Reference (User)
Branch	Choice
Email	String
Phone Number	String
Description	Multi-line String

7. Security Requirements

The Read ACL should verify:

- Admin users receive full access.
- Users with the bb1 role can access only EEE records.

- Other users cannot view restricted records.

Create, Write, and Delete permissions should be controlled using bb2, bb3, and bb4 roles.

8. Expected Output

The system should successfully protect student records using role-based security while allowing authorized users to perform only their permitted operations.

Phase 3 – Project Design Phase

1. Introduction

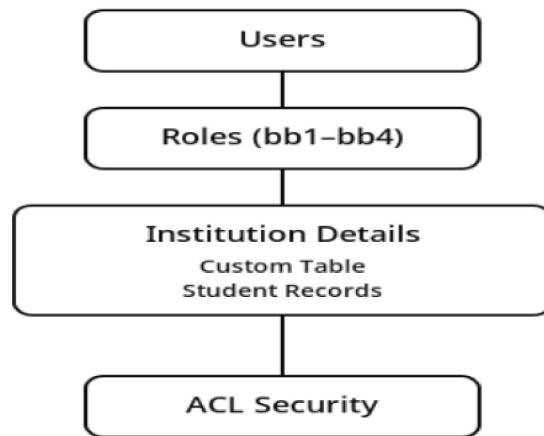
The Project Design phase defines how the ServiceNow application is structured before implementation. It explains the table design, field structure, user roles, ACL workflow, and system architecture used to secure the Institution Details records.

2. System Architecture

The application consists of four main components:

- Users (Admin and EEE User)
- Roles (bb1, bb2, bb3, bb4)
- Institution Details custom table
- Access Control Lists (ACLs)

Architecture Diagram



3. Table Design

Custom Table

Property	Value
Label	Institution Details
Name	u_institution_details
Extends	False

4. Field Design

Field Name	Data Type	Purpose
Student Roll Number	Auto Number	Unique ID
Student Name	Reference (User)	Student details
Faculty Name	Reference (User)	Faculty details
Branch	Choice	ECE, EEE, CSE
Email	String	Contact
Phone Number	String	Contact
Description	Multi-line String	Additional details

5. User and Role Design

User	Role	Permission
Admin	admin	Full access
EEE User	bb1	Read EEE records
EEE User	bb2	Create records
EEE User	bb3	Edit records
EEE User	bb4	Delete records

6. ACL Design

Read ACL

- Type: Record
- Operation: Read
- Table: u_institution_details
- Role: bb1
- Data Condition: Branch = EEE

Create ACL

- Role: bb2

Write ACL

- Role: bb3

Delete ACL

- Role: bb4

7. Expected Design Outcome

The designed system ensures that access is controlled through user roles and the Branch field. The ACL structure provides secure record-level protection while maintaining full administrative control over the Institution Details application.

Phase 4 – Project Planning Phase

1. Introduction

Project Planning is the process of organizing tasks, assigning responsibilities, identifying required resources, and creating a timeline before development. This phase ensures that the project is completed systematically within the allotted time.

2. Project Objectives

- Create a secure ServiceNow application.
- Implement Script-Controlled ACLs.
- Restrict record access based on user roles and the Branch field.
- Verify Read, Create, Write, and Delete permissions.
- Complete all project phases successfully.

3. Project Scope

The project includes:

- User creation
- Role creation (bb1, bb2, bb3, bb4)
- Institution Details table creation
- Field configuration
- ACL implementation
- Testing through impersonation
- Documentation and demonstration

4. Work Breakdown Structure (WBS)

Task ID	Task	Status
1	Create User	Completed
2	Create Roles	Completed
3	Create Custom Table	Completed
4	Add Fields	Completed
5	Create Records	Completed
6	Configure Read ACL	Completed
7	Configure Create ACL	Completed
8	Configure Write ACL	Completed
9	Configure Delete ACL	Completed
10	Testing	Completed
11	Documentation	Completed
12	Demo Preparation	Completed

5. Timeline

Phase	Time
User & Role Creation	20 min
Table Creation	25 min
Record Creation	15 min
ACL Configuration	60 min
Testing	20 min
Documentation	30 min
Demo Preparation	10 min

Total Duration: 3 Hours

6. Resources Required

Software

- ServiceNow Developer Instance
- Web Browser
- GitHub

Hardware

- Laptop/Desktop
- Internet Connection

7. Risk Management

Risk	Solution
Wrong role assignment	Verify assigned roles
ACL misconfiguration	Test using impersonation
Missing records	Create sample records
Permission errors	Review ACL conditions

8. Team Responsibilities

Member	Responsibility
Member 1	User & Role Creation
Member 2	Table & Field Creation
Member 3	ACL Configuration
Member 4	Testing & Documentation

9. Expected Deliverables

- Institution Details application
- Custom table
- Four ACLs (Read, Create, Write, Delete)
- Testing results
- GitHub repository
- Demo video

Phase 5 – Project Development Phase

1. Introduction

The Project Development phase focuses on implementing the planned features in the ServiceNow platform. During this phase, users, roles, custom tables, fields, and Access Control Lists (ACLs) were created and configured to enforce secure record-level access based on the user's role and the Branch field.

2. Development Environment

Component	Details
Platform	ServiceNow Developer Instance
Language	JavaScript (ACL Script)
Security	Access Control Lists (ACLs)

Database	ServiceNow Custom Table
Browser	Google Chrome

3. User Creation

A test user named EEE User was created with the following details.

Field	Value
User ID	EEE User
First Name	EEE
Last Name	User
Email	eeeuser@gmail.com

4. Role Creation

Four custom roles were created.

Role	Purpose
bb1	Read Access

bb2	Create Access
bb3	Write/Edit Access
bb4	Delete Access

All four roles were assigned to the EEE User.

5. Custom Table Development

A custom table named Institution Details (u_institution_details) was created.

Table Fields

Field Name	Data Type
Student Roll Number	Auto Number
Student Name	Reference (User)
Faculty Name	Reference (User)
Branch	Choice
Email	String
Phone Number	String
Description	Multi-line String

dev395038.service-now.com/now/nav/ui/classic/params/target/sys_db_object.do%3Fsys_id%3D2aef8930c3630f10949353f1b4013106%26sysparm_record_target%3Dsys_db_...

servicenow

AllFavoritesHistoryAdmin

Table - Institution Details

Search

DeleteUpdateDelete All Records

Table

Institution Details

DeleteUpdateDelete All Records

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label

Institution Details

Application

Global

* Name

u_institution_details

Remote Table

ColumnsControlsApplication Access

Table Columns

for text

Search

1 to 13 of 13

New

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Updated by	String	(empty)	40		false
Updates	Integer	(empty)	40		false
Sys ID	Sys ID (GUID)	(empty)	32		false
Student Roll Number	String	(empty)	40		false
Faculty Name	Reference	User	32		false

servicenow

AllFavoritesHistoryAdmin

Table - Institution Details

Search

DeleteUpdateDelete All Records

Table

Institution Details

DeleteUpdateDelete All Records

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Updated by	String	(empty)	40		false
Updates	Integer	(empty)	40		false
Sys ID	Sys ID (GUID)	(empty)	32		false
Student Roll Number	String	(empty)	40		false
Faculty Name	Reference	User	32		false
Updated	Date/Time	(empty)	40		false
Student Name	Reference	User	32		false
Branch	Choice	(empty)	40		false
Created by	String	(empty)	40		false
Email	String	(empty)	40		false
Description	String	(empty)	40		false
Phone Number	String	(empty)	40		false
Created	Date/Time	(empty)	40		false
Insert a new row...					

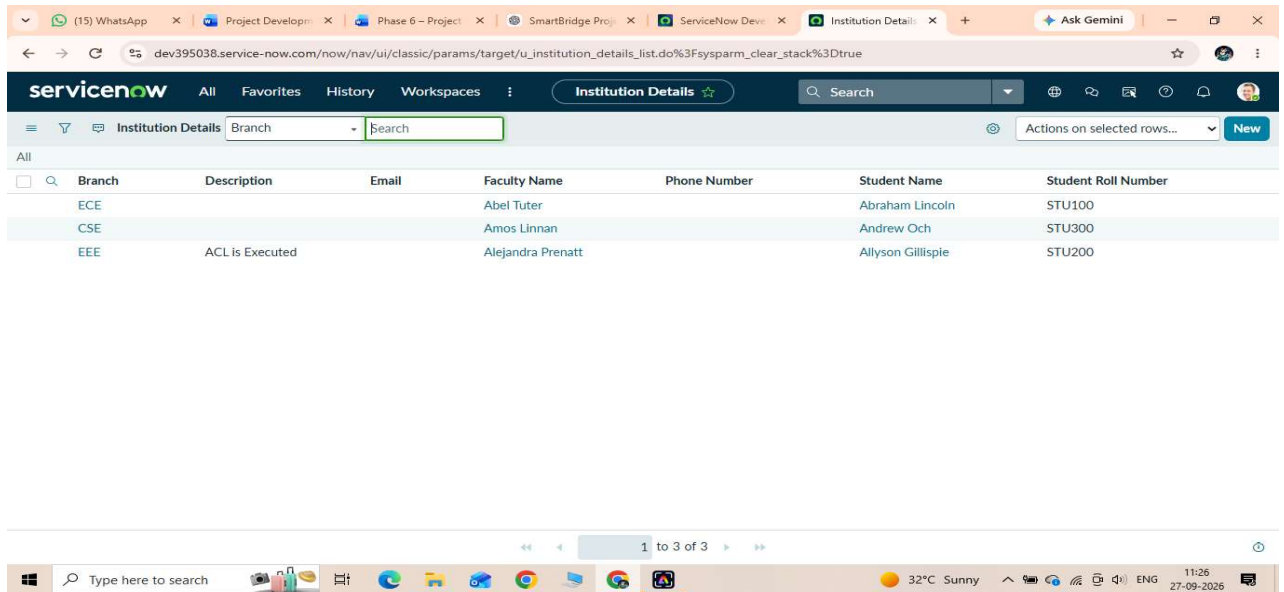
Name	Decision Type	Operation	Type	Active	Updated by	Updated
u_institution_details	Allow If	delete	record	true	admin	2026-09-22 20:04:05
u_institution_details	Allow If	write	record	true	admin	2026-09-22 20:02:40
u_institution_details	Allow If	create	record	true	admin	2026-09-22 20:00:32
u_institution_details	Allow If	read	record	true	admin	2026-09-22 19:58:04
u_institution_details	Allow If	delete	record	true	admin	2026-09-22 19:37:29
u_institution_details	Allow If	write	record	true	admin	2026-09-22 19:37:29
u_institution_details	Allow If	create	record	true	admin	2026-09-22 19:37:29
u_institution_details	Allow If	read	record	true	admin	2026-09-22 19:37:29

6. Sample Record Creation

Multiple student records were created using different branch values.

Student	Branch
Student 1	EEE
Student 2	ECE
Student 3	CSE

These records were used for testing ACL behavior.



7. ACL Implementation

Read ACL

Property	Value
Type	Record
Operation	Read
Table	u_institution_details
Role	bb1
Condition	Branch = EEE

Script Used

JavaScript

```
(function () {  
    if (gs.hasRole('admin')) {  
        return true;  
    }  
  
    if (gs.hasRole('bb1')) {  
        return true;  
    }  
  
    return false;  
})();
```

The screenshot displays the ServiceNow interface for configuring an Access Control rule. The rule is named 'u_institution_details' and is set to 'read' operation. The 'Type' is 'record' and the 'Application' is 'Global'. The 'Active' checkbox is checked, and the 'Advanced' checkbox is also checked. The 'Decision Type' is set to 'Allow If'. The 'Protection policy' is set to '-- None --'. The 'Name' is 'u_institution_details' and the 'Description' is 'Allow read for records in u_institution_details, for users with role bb1, and if the ACL script returns true, and if the ACL condition (u_branch=EEE^EQ) evaluates to true.'

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

1 to 1 of 1

Advanced Condition

Controlled by References

Script ☐ Turn on ECMAScript 2021 (ES12) mode

```
1 (function () {
2     // Allow admin users full access
3     if (gs.hasRole('admin')) {
4         return true;
5     }
6     // Allow only EEE branch users to see EEE records
7     if (gs.hasRole('bb1')) {
8         return true;
9     }
10    // Deny access for all others
11    return false;
12 })();
13
```

Related Links

- [Show ACL Execution Plan](#)
- [Run Point Scan](#)

Create ACL

- Role: bb2
- Operation: Create

Write ACL

- Role: bb3
- Operation: Write

Delete ACL

- Role: bb4
- Operation: Delete

8. Development Workflow

The implementation followed these steps:

1. Created the EEE User.
2. Created four custom roles.
3. Assigned roles to the user.
4. Created the Institution Details table.
5. Added all required fields.
6. Inserted sample records.
7. Configured Read ACL with JavaScript.
8. Created Create, Write, and Delete ACLs.
9. Verified permissions using impersonation.

9. Development Challenges

Challenge	Solution
Role assignment	Verified assigned roles
ACL configuration	Tested using impersonation

Record visibility	Checked Branch condition
Permission errors	Reviewed ACL settings

10. Development Outcome

The development phase successfully implemented a secure ServiceNow application where administrators retained full access, users with bb1 could view EEE records, bb2 enabled record creation, bb3 allowed editing, and bb4 permitted deletion. The application achieved role-based record security through Script-Controlled ACLs.

Phase 6 – Project Testing

1. Introduction

The Project Testing phase verifies that the ServiceNow application works correctly after development. Testing was performed using Impersonate User to confirm that the Read, Create, Write, and Delete ACLs provide the correct permissions for different users.

2. Testing Objectives

- Verify Read ACL.
- Verify Create ACL.
- Verify Write ACL.
- Verify Delete ACL.
- Confirm Admin has full access.

3. Testing Environment

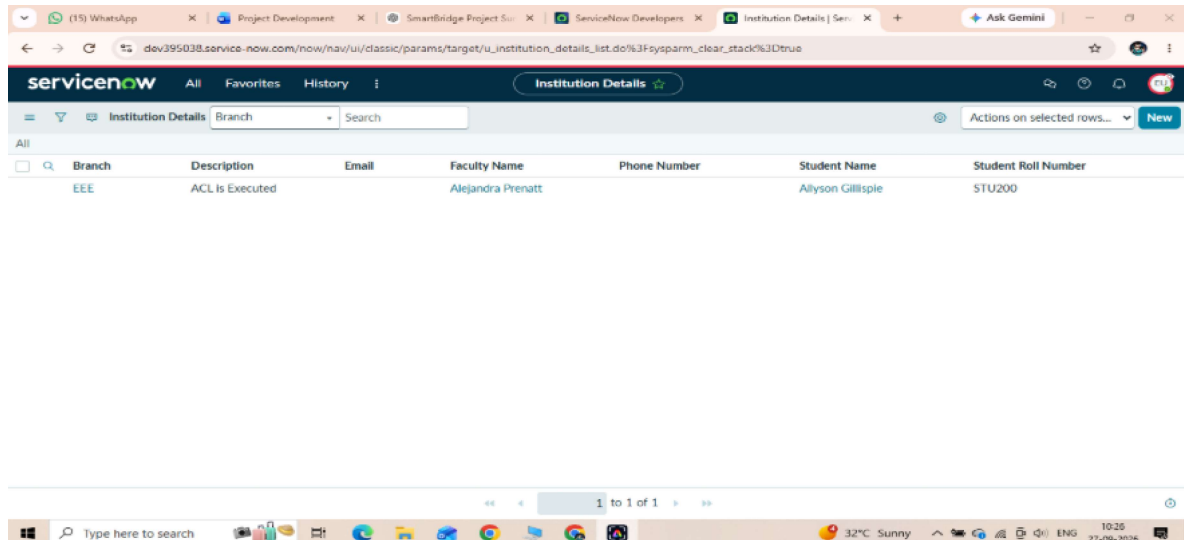
Component	Details
Platform	ServiceNow Developer Instance
Table	u_institution_details
Test User	EEE User
Roles	bb1, bb2, bb3, bb4

4. Test Cases

Test	Expected Result	Status
Read ACL	Only EEE records visible	Pass
User without role	No records visible	Pass
Admin	All records visible	Pass
Create ACL	New button visible	Pass
Write ACL	EEE records editable	Pass
Delete ACL	EEE records deletable	Pass

5. Verification Screenshots

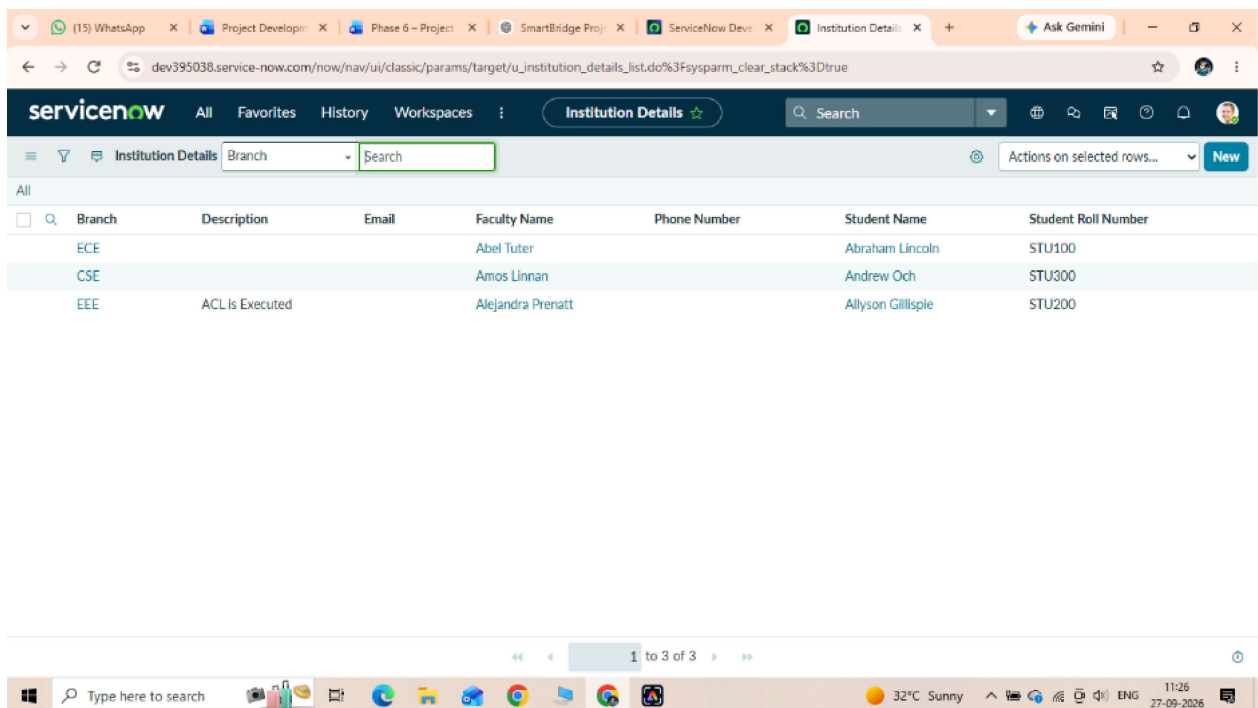
Figure 1 – EEE User Viewing Only EEE Records



The screenshot shows a ServiceNow web interface. The browser tabs include WhatsApp, Project Development, SmartBridge Project, ServiceNow Developers, and Institution Details. The URL is dev395038.service-now.com/now/nav/ui/classic/params/target/u_institution_details_list.do%3Fsysparm_clear_stack%3Dtrue. The page title is "Institution Details". The table has columns: Branch, Description, Email, Faculty Name, Phone Number, Student Name, and Student Roll Number. Only one record is visible for the EEE branch.

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE	ACL Is Executed		Alejandra Prenatt		Allyson Gillispie	STU200

Figure 2 – Admin Viewing All Records



The screenshot shows the same ServiceNow web interface as Figure 1, but from an Admin user's perspective. The browser tabs include WhatsApp, Project Development, Phase 6 - Project, SmartBridge Proj, ServiceNow Dev, and Institution Details. The URL is the same. The page title is "Institution Details". The table has columns: Branch, Description, Email, Faculty Name, Phone Number, Student Name, and Student Roll Number. Three records are visible: ECE, CSE, and EEE.

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
ECE			Abel Tuter		Abraham Lincoln	STU100
CSE			Amos Linnan		Andrew Och	STU300
EEE	ACL Is Executed		Alejandra Prenatt		Allyson Gillispie	STU200

Figure 3 – Create ACL (New Button Visible)

Access Control - u_institution_details

Type: record

Operation: create

Application: Global

Active: ☒

Admin overrides: ☒

Decision Type: Allow if

Protection policy: -- None --

Name: u_institution_details

Description: Allow create for records in u_institution_details, for users with role bb2.

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Institution Details

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE	ACL is Executed		Alejandra Prenatt		Allyson Gillispie	STU200

New

Figure 4 – Write ACL (Edit Record)

dev395038.service-now.com/now/nav/ui/classic/params/target/sys_security_acl.do%3Fsys_id%3D6e655df8c3630f10949353f1b40131e2%26sysparm_record_target%3Dsys_se...

servicenow All Favorites History Access Control - u_institution_details Search

Access Control
u_institution_details

Type record Application Global
Operation write Active ☒
Admin overrides ☒ Advanced ☐
Decision Type Allow If
Protection policy -- None --
Name u_institution_details
Description Allow write for records in u_institution_details, for users with role bb3.

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.
1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.
[More Info](#)

Requires role 1 to 1 of 1

(15) WhatsApp X Project Develop... X Phase 6 - Project X SmartBridge Proj... X ServiceNow Dev... X Created 2026-09-27 X Ask Gemini

dev395038.service-now.com/now/nav/ui/classic/params/target/u_institution_details.do%3Fsys_id%3D6d03ddb4c3630f10949353f1b40131f%26sysparm_record_target%3Du...

servicenow All Favorites History Institution Details - Created 2026-0... Update Delete

Institution Details 19:51:36

Save
Insert
Insert and Stay
Export
Create Favorite
Copy URL
Reload form

STU200
PDF natt
Allyson Gillispie
EEE

Email
Description ACL is Executed
Phone Number

Update Delete

Figure 5 – Delete ACL (Delete Option)

Access Control - u_institution_details

Type: record

Operation: delete

Application: Global

Active: ☒

Admin overrides: ☒

Decision Type: Allow if

Protection policy: -- None --

Name: u_institution_details

Description: Allow delete for records in u_institution_details, for users with role bb4.

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Figure 6 – Final Verification

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE	ACL is Executed		Alejandra Prenatt		Allyson Gillispie	STU200

6. Testing Summary

- bb1 – View only EEE branch records.
- bb2 – Create new records.

- bb3 – Edit EEE records.
- bb4 – Delete EEE records.
- Admin – Full access to all records.

PHASE 7 – PROJECT DOCUMENTATION

7.1 Introduction

Project Documentation is the process of recording all important information about the project, including the technologies used, implementation details, testing results, and project outcomes. Proper documentation helps future users understand how the ServiceNow application was developed and how the ACLs were configured.

7.2 Technologies Used

Technology	Purpose
ServiceNow	Platform for application development
JavaScript	Script-Controlled Read ACL
Access Control Lists (ACLs)	Record-level security
Role-Based Access Control	Permission management
GitHub	Project version control
Google Drive	Demo video sharing

7.3 Project Features

The application includes the following features:

- Custom Institution Details table.
- User creation and role assignment.
- Four custom roles (bb1, bb2, bb3, bb4).
- Read ACL with JavaScript.
- Create ACL.
- Write ACL.
- Delete ACL.
- Impersonation-based testing.

7.4 Security Implementation

The project implements role-based security by combining user roles with record conditions.

ACL	Purpose
Read	View only EEE records
Create	Allow new records
Write	Edit records
Delete	Delete records
Admin	Full access

7.5 Advantages

- Protects confidential student records.
- Prevents unauthorized access.
- Demonstrates practical ServiceNow security.
- Supports secure record management.
- Easy to manage using custom roles.

7.6 Limitations

- The project is limited to the EEE branch condition.
- Requires proper role assignment.
- Depends on ServiceNow permissions.
- Additional conditions may require extra scripting.

7.7 Future Enhancements

The application can be improved by adding:

- Department-wise dynamic permissions.
- Automatic role assignment.
- Email notifications.
- Audit logging.
- Approval workflows.

7.8 Documentation Screenshots

Add 2–3 screenshots here:

1. Institution Details Table

* Label Institution Details
* Name u_institution_details
Extends table

Application Global
Create module ☒
Create mobile module ☒
Add module to menu -- Create new --
New menu name
Remote Table ☐

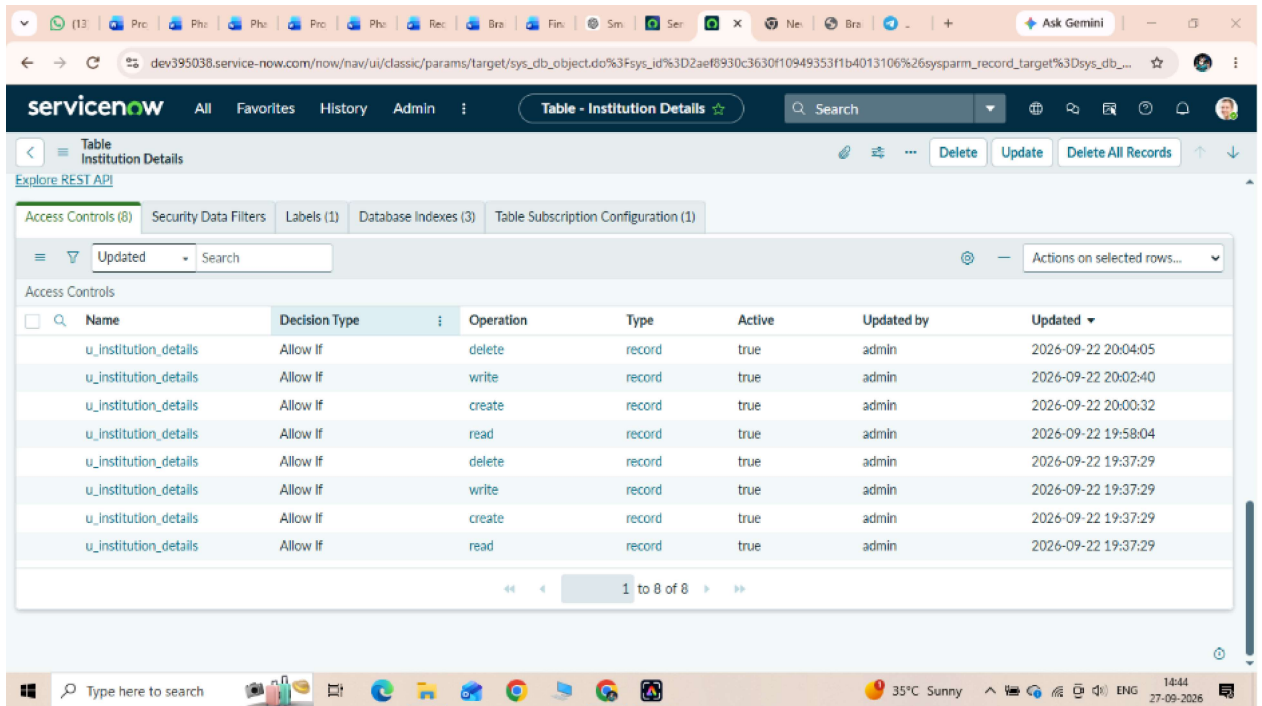
Columns Controls Application Access

Table Columns for text Search

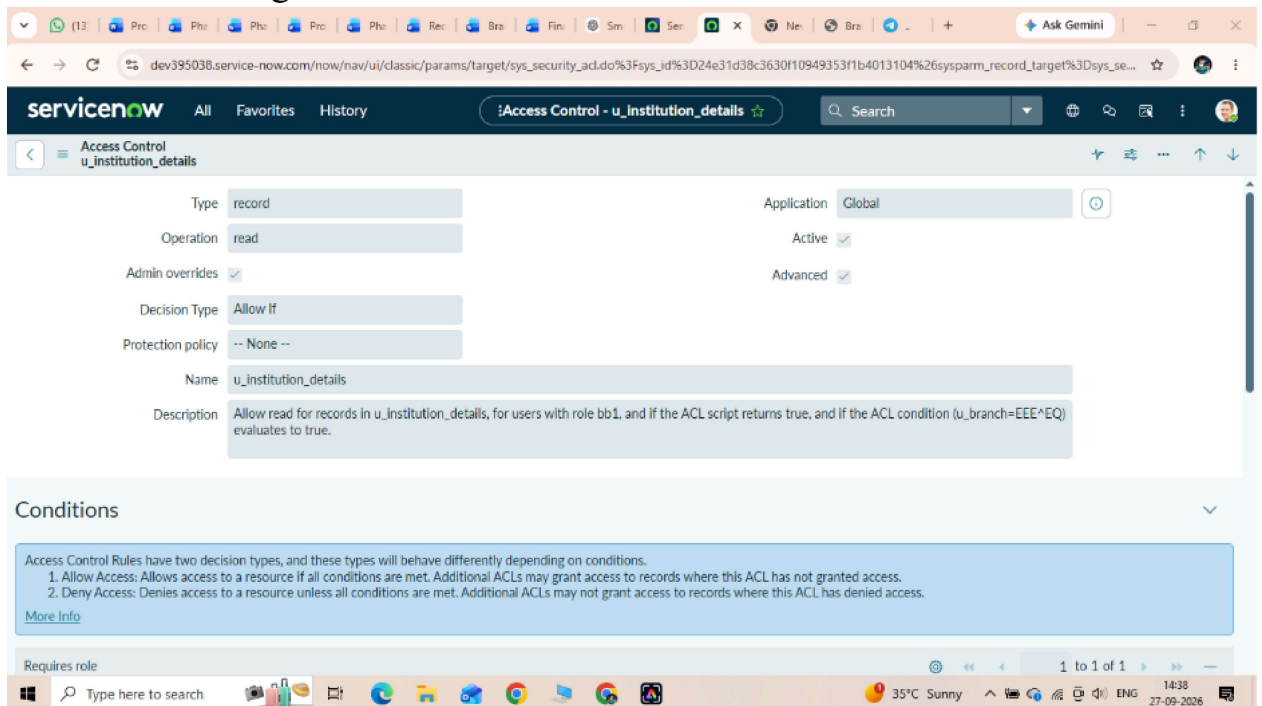
Column label	Type	Reference	Max length	Default value	Display
Updated by	String	(empty)		40	false
Updates	Integer	(empty)		40	false
Sys ID	Sys ID (GUID)	(empty)		32	false

Table Institution Details

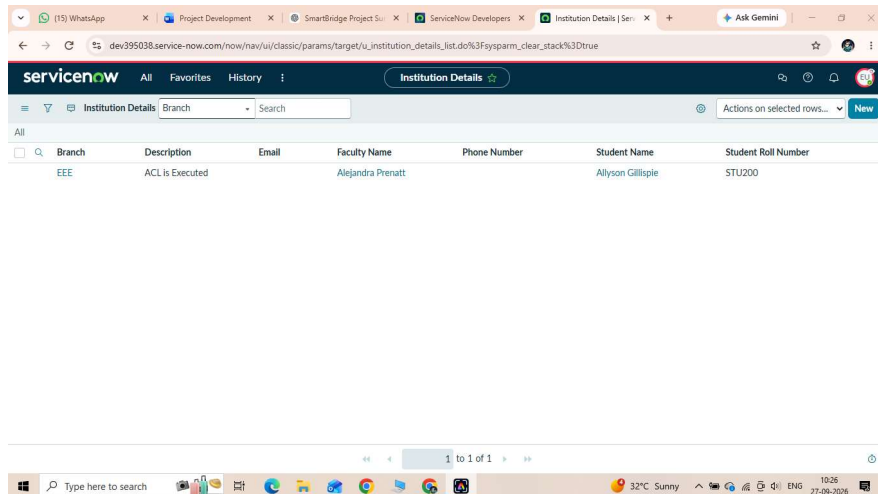
Column label	Type	Reference	Max length	Default value	Display
Updated by	String	(empty)		40	false
Updates	Integer	(empty)		40	false
Sys ID	Sys ID (GUID)	(empty)		32	false
Student Roll Number	String	(empty)		40	false
Faculty Name	Reference	User		32	false
Updated	Date/Time	(empty)		40	false
Student Name	Reference	User		32	false
Branch	Choice	(empty)		40	false
Created by	String	(empty)		40	false
Email	String	(empty)		40	false
Description	String	(empty)		40	false
Phone Number	String	(empty)		40	false
Created	Date/Time	(empty)		40	false



2. Read ACL Configuration



3. Final Working Screen (EEE User or Admin)



Phase 8 – Project Demonstration

8.1 Introduction

The Project Demonstration phase presents the completed ServiceNow application by demonstrating each milestone of the implementation. The demonstration verifies that Script-Controlled Access Control Lists (ACLs) successfully restrict record access based on user roles and the Branch field value while administrators retain full access.

Milestone 1 – Creation of Users and Roles

User Creation

1. Log in to the ServiceNow instance with Admin access.
2. Navigate to User Administration → Users.
3. Click New.
4. Create the EEE User with the following details:
 - a. User ID: EEE User
 - b. First Name: EEE
 - c. Last Name: User
 - d. Email: eeeuser@gmail.com

5. Save the record.

Screenshot: EEE User Record.

The screenshot shows the ServiceNow User Administration interface for a user named 'EEE User'. The page is divided into two main sections: user details on the left and user preferences on the right. The user details section includes fields for User ID (EEE User), First name (EEE), Last name (User), Title, Department, Password needs reset, Locked out, Active (checked), Identity type (Human), and Internal Integration User. The user preferences section includes fields for Email (eeeuser@gmail.com), Language (None), Calendar integration (Outlook), Time zone (System (America/Los Angeles)), Date format (System (2026-08-19)), Business phone, Mobile phone, and Photo (Click to add...). At the bottom of the page, there are buttons for 'Update', 'Set Password', and 'Delete', and a 'Related Links' section with links for 'View linked accounts', 'View Subscriptions', and 'Reset a password'. The browser's address bar shows the URL: dev395038.service-now.com/now/nav/ui/classic/params/target/sys_user.do%3Fsys_id%3DDaed853cc3230f10949353f1b40131cf%26sysparm_record_target%3Dsys_user%26s...

Role Creation

1. Navigate to User Administration → Roles.
2. Create the following roles:
 - a. bb1
 - b. bb2
 - c. bb3
 - d. bb4
3. Assign all four roles to EEE User.

Screenshot: Role Creation and Role Assignment.

dev395038.service-now.com/now/nav/ui/classic/params/target/sys_user_role.do%3Fsys_id%3Db2be81bcc3230f10949353f1b4013125%26sysparm_record_target%3Dsys_user...

servicenowAllFavoritesHistoryWorkspacesRole - bb1Search

Rolebb1

Namebb1ApplicationGlobalElevated privilege

Description

UpdateDelete

Contains RolesApplications with RoleModules with RoleCustom Tables

for text SearchNewEdit

Role = bb1

Contains

dev395038.service-now.com/now/nav/ui/classic/params/target/sys_user_role_list.do%3Fsysparm_first_row%3D1%26sysparm_query%3DGOtoName%253e%253dbb%26syspa...

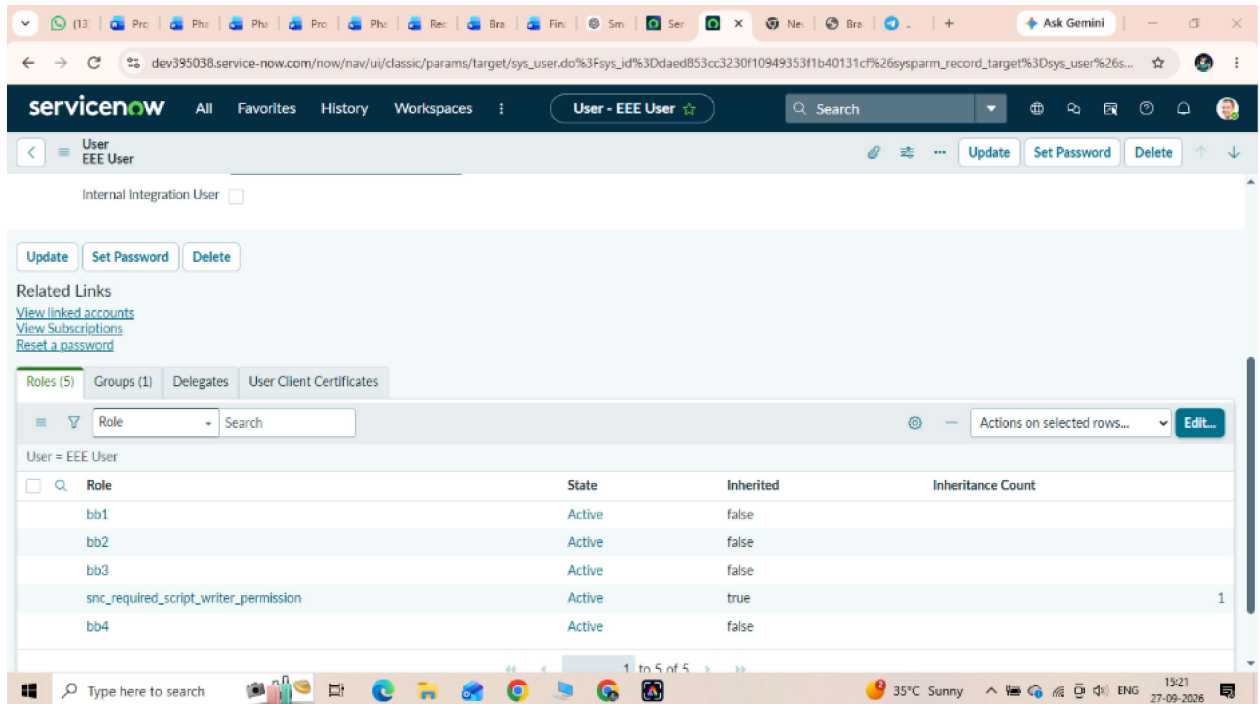
servicenowAllFavoritesHistoryWorkspacesRolesSearch

RolesName Search

All > Name >= bb

Name	Description	Elevated privilege
Search	Search	Search
bb1		false
bb2		false
bb3		false
bb4		false
bsm_legacy	Only users with the 'bsm_legacy' will see the module BSM Legacy within Dependencies View	false
bsm_legacy_admin	Only users with the 'bsm_legacy_admin' will see the module BSM Legacy within Dependencies View	false
business_calendar_admin	Access to configure and manage plugins and features related to business calendars.	false
business_process_admin		false
business_process_manager		false
business_process_user		false
business_relationship_manager	Person(s) responsible for both business strategy and business value in an effort to advance business objectives.	false
business_rule_admin	Can manage Business Rules	false

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Milestone 2 – Institution Details Table Creation

1. Navigate to System Definition → Tables.
2. Create the custom table Institution Details (u_institution_details).
3. Add the following fields:
 - a. Student Roll Number (Auto Number)
 - b. Student Name (Reference)
 - c. Faculty Name (Reference)
 - d. Branch (Choice: ECE, EEE, CSE)
 - e. Email (String)
 - f. Phone Number (String)
 - g. Description (Multi-line String)
4. Save the table.
5. Create sample records for ECE, EEE, and CSE branches.

Screenshot: Institution Details Table and Sample Records.

dev395038.service-now.com/now/nav/ui/classic/params/target/sys_db_object_list.do%3Fsysparm_userpref_module%3D7e7ca89ac0a8000901594ba32405461%26sysparm_q...

serviceNow

Tables

Search

Table Institution Details

* Label

Institution Details

* Name

u_institution_details

Extends table

Application

Global

Create module

☒

Create mobile module

☒

Add module to menu

-- Create new --

New menu name

Remote Table

☐

Columns

Controls

Application Access

Table Columns

for text

Search

New

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Updated by	String	(empty)		40	false
Updates	Integer	(empty)		40	false
Sys ID	Sys ID (GUID)	(empty)		32	false

dev395038.service-now.com/now/nav/ui/classic/params/target/sys_db_object.do%3Fsys_id%3D2aef8930c3630f10949353f1b4013106%26sysparm_record_target%3Dsys_db_...

serviceNow

Table - Institution Details

Search

Table Institution Details

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Updated by	String	(empty)		40	false
Updates	Integer	(empty)		40	false
Sys ID	Sys ID (GUID)	(empty)		32	false
Student Roll Number	String	(empty)		40	false
Faculty Name	Reference	User		32	false
Updated	Date/Time	(empty)		40	false
Student Name	Reference	User		32	false
Branch	Choice	(empty)		40	false
Created by	String	(empty)		40	false
Email	String	(empty)		40	false
Description	String	(empty)		40	false
Phone Number	String	(empty)		40	false
Created	Date/Time	(empty)		40	false
Insert a new row...					

The screenshot shows the ServiceNow interface for the 'Access Controls' table. The table has 8 records, all with the name 'u_institution_details'. The columns are: Name, Decision Type, Operation, Type, Active, Updated by, and Updated. The records show various operations (delete, write, create, read) being performed on the 'u_institution_details' record, all with a decision type of 'Allow If' and an active status of 'true'. The records were updated by 'admin' at various times on 2026-09-22.

Name	Decision Type	Operation	Type	Active	Updated by	Updated
u_institution_details	Allow If	delete	record	true	admin	2026-09-22 20:04:05
u_institution_details	Allow If	write	record	true	admin	2026-09-22 20:02:40
u_institution_details	Allow If	create	record	true	admin	2026-09-22 20:00:32
u_institution_details	Allow If	read	record	true	admin	2026-09-22 19:58:04
u_institution_details	Allow If	delete	record	true	admin	2026-09-22 19:37:29
u_institution_details	Allow If	write	record	true	admin	2026-09-22 19:37:29
u_institution_details	Allow If	create	record	true	admin	2026-09-22 19:37:29
u_institution_details	Allow If	read	record	true	admin	2026-09-22 19:37:29

Milestone 3 – Read ACL Demonstration

1. Navigate to System Security → Access Control (ACL).
2. Elevate the Security_admin role.
3. Create a Read ACL with:
 - a. Type: Record
 - b. Operation: Read
 - c. Name: u_institution_details
 - d. Requires Role: bb1
 - e. Data Condition: Branch = EEE
4. Add the JavaScript script.
5. Save the ACL.
6. Impersonate EEE User and open u_institution_details.list.
7. Verify that only EEE records are visible.
8. Impersonate Admin and verify that all records are visible.

Screenshots:

- Read ACL

dev395038.service-now.com/now/nav/ui/classic/params/target/sys_security_acl.do%3Fsys_id%3D24e31d38c3630f10949353f1b4013104%26sysparm_record_target%3Ds...

servicenow All Favorites History Access Control - u_institution_details Search

Access Control
u_institution_details

Type record Application Global

Operation read Active ☒

Admin overrides ☒ Advanced ☒

Decision Type Allow If

Protection policy -- None --

Name u_institution_details

Description Allow read for records in u_institution_details, for users with role bb1, and if the ACL script returns true, and if the ACL condition (u_branch=EEE^EQ) evaluates to true.

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role 1 to 1 of 1

- EEE User View

dev395038.service-now.com/now/nav/ui/classic/params/target/u_institution_details_list.do%3Fsysparm_clear_stack%3Dtrue

servicenow All Favorites History Institution Details Search

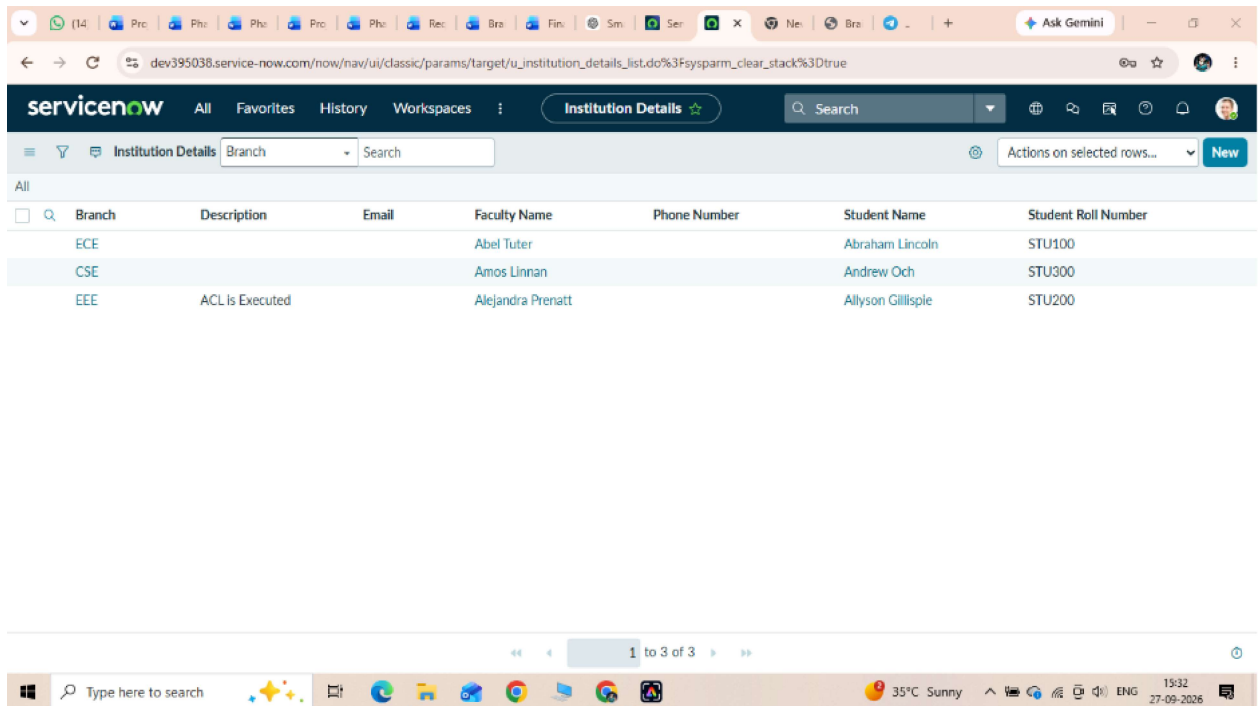
Actions on selected rows... New

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE	ACL Is Executed		Alejandra Prenatt		Allyson Gillisple	STU200

1 to 1 of 1

32°C Sunny 10:28 27-09-2026

- Admin View



Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
ECE			Abel Tuter		Abraham Lincoln	STU100
CSE			Amos Linnian		Andrew Och	STU300
EEE	ACL is Executed		Alejandra Prenatt		Allyson Gillisple	STU200

Milestone 4 – Create ACL Demonstration

1. Create a Create ACL.
2. Configure:
 - a. Type: Record
 - b. Operation: Create
 - c. Name: u_institution_details
 - d. Requires Role: bb2
3. Save the ACL.
4. Impersonate EEE User.
5. Verify that the New button appears.

Screenshot: New Button.

The screenshot shows the ServiceNow interface for the 'Institution Details' list. The table has the following data:

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE	ACL is Executed		Alejandra Prenatt		Allyson Gillispie	STU200

Milestone 5 – Write ACL Demonstration

1. Create a Write ACL.
2. Configure:
 - a. Type: Record
 - b. Operation: Write
 - c. Name: u_institution_details
 - d. Requires Role: bb3
3. Save the ACL.
4. Open an EEE record.
5. Edit the record and save it.

Screenshot: Edit Record.

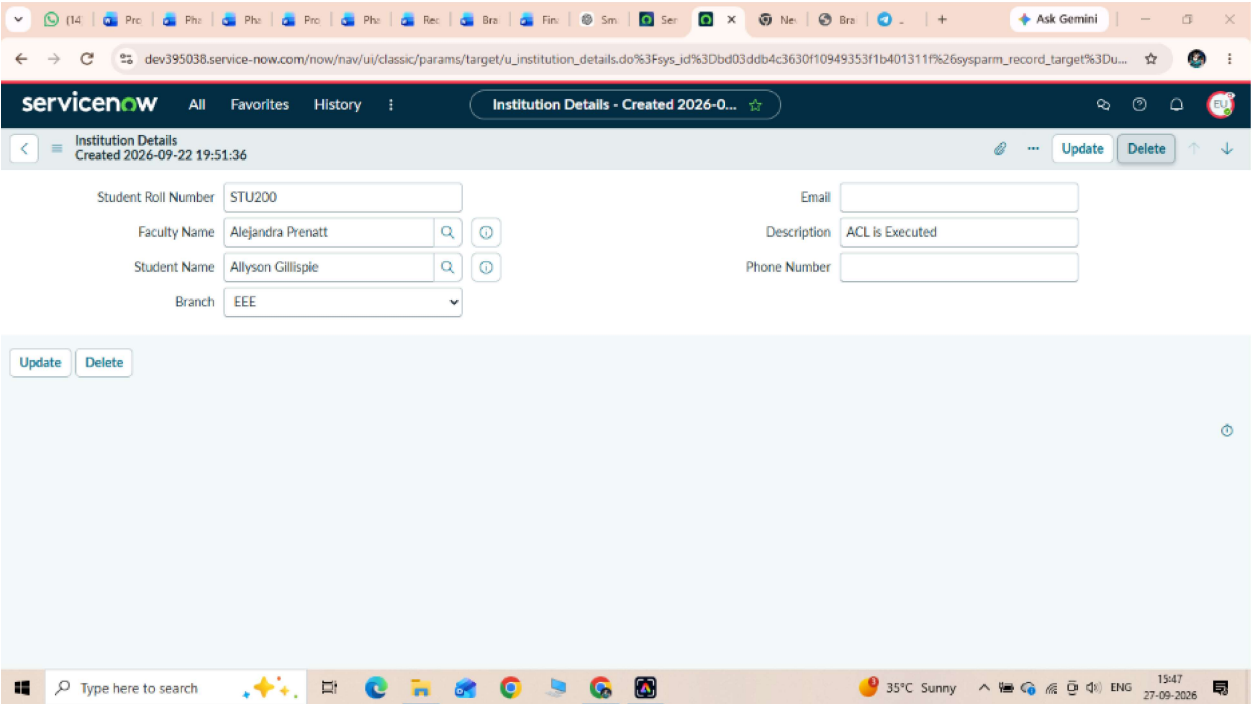
The screenshot shows the ServiceNow user interface for editing an 'Institution Details' record. The browser address bar shows a URL from dev395038.service-now.com. The page header includes the ServiceNow logo and navigation tabs like 'All', 'Favorites', and 'History'. The record title is 'Institution Details - Created 2026-0-...' with a star icon. Below the title, the record is identified as 'Institution Details' created on '2026-09-22 19:51:36'. There are 'Update' and 'Delete' buttons in the top right. The form contains several fields: 'Student Roll Number' (STU200), 'Faculty Name' (Alejandra Prenatt), 'Student Name' (Allyson Gillispie), 'Branch' (EEE), 'Email', 'Description' (ACL is Executed), and 'Phone Number'. Search and refresh icons are next to the Faculty Name and Student Name fields. At the bottom of the form area, there are 'Update' and 'Delete' buttons. The Windows taskbar at the bottom shows the search bar, task view, and system tray with weather and date information.

Student Roll Number	STU200	Email	
Faculty Name	Alejandra Prenatt	Description	ACL is Executed
Student Name	Allyson Gillispie	Phone Number	
Branch	EEE		

Milestone 6 – Delete ACL Demonstration

1. Create a Delete ACL.
2. Configure:
 - a. Type: Record
 - b. Operation: Delete
 - c. Name: u_institution_details
 - d. Requires Role: bb4
3. Save the ACL.
4. Open an EEE record.
5. Verify that the Delete option is available.

Screenshot: Delete Option.



Final Verification

User	Permission
Admin	View, Create, Edit, Delete all records
bb1	View only EEE records
bb2	Create new records
bb3	Edit EEE records
bb4	Delete EEE records

Outcome

The demonstration successfully verified all six milestones. The ServiceNow application correctly enforced Read, Create, Write, and Delete permissions using Script-Controlled ACLs, ensuring that EEE branch records were accessible only to authorized users while administrators retained complete control over the system.

Conclusion

The Script-Controlled ACL – Restrict Record Access Based on Field Value project was successfully completed using the ServiceNow platform. The project demonstrated how Access Control Lists (ACLs) can be used to secure records by combining user roles with the Branch field value. A custom Institution Details table was created, and users, roles, and Read, Create, Write, and Delete ACLs were configured to enforce role-based access.

The testing results confirmed that users with the appropriate roles could perform only their permitted operations, while administrators retained full access to all records. This project improved our understanding of role-based security, record-level access control, ACL scripting, and ServiceNow application development, providing practical experience in implementing secure access management.

